



Analyzing SpaceX Launch Success Factors

Objectives

- To analyze the factors that influence the success of the first stage landing.
- Identify launch success and failure rates.
- Develop a predictive model to predict the outcome of launches.



Methodology

Data collection using SpaceX API and Wikipedia web scraping.

Exploratory data analysis (EDA) with visualizations and SQL.

Creation of an interactive dashboard with Plotly Dash.

Classification models for predictive analytics.



Deliverables

Completed Notebooks: Data Wrangling, EDA, SQL,
and Machine Learning.

<https://github.com/4sur/IBMDDataScienceCapstone>

Significance of the study:

```
# Plot a line chart with x axis to be the extracted year and y axis to be the success rate
```

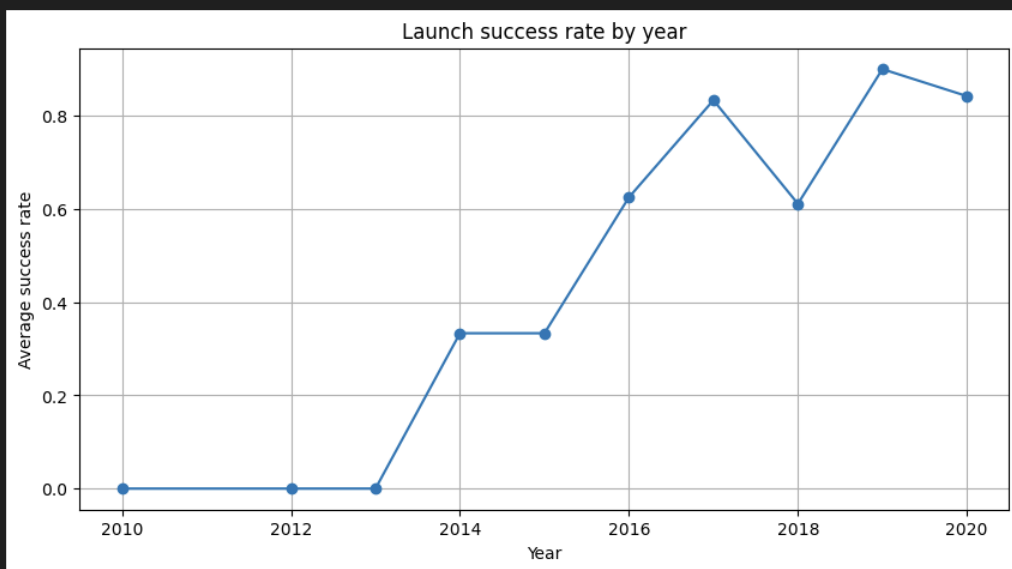
```
df['Year'] = pd.to_datetime(df['Date']).dt.year
```

```
success_rate = df.groupby('Year')['Class'].mean()
```

```
plt.figure(figsize=(10, 5))  
plt.plot(success_rate.index, success_rate.values, marker='o')  
plt.title('Launch success rate by year')  
plt.xlabel('Year')  
plt.ylabel('Average success rate')  
plt.grid(True)  
plt.show()
```

[28]

...

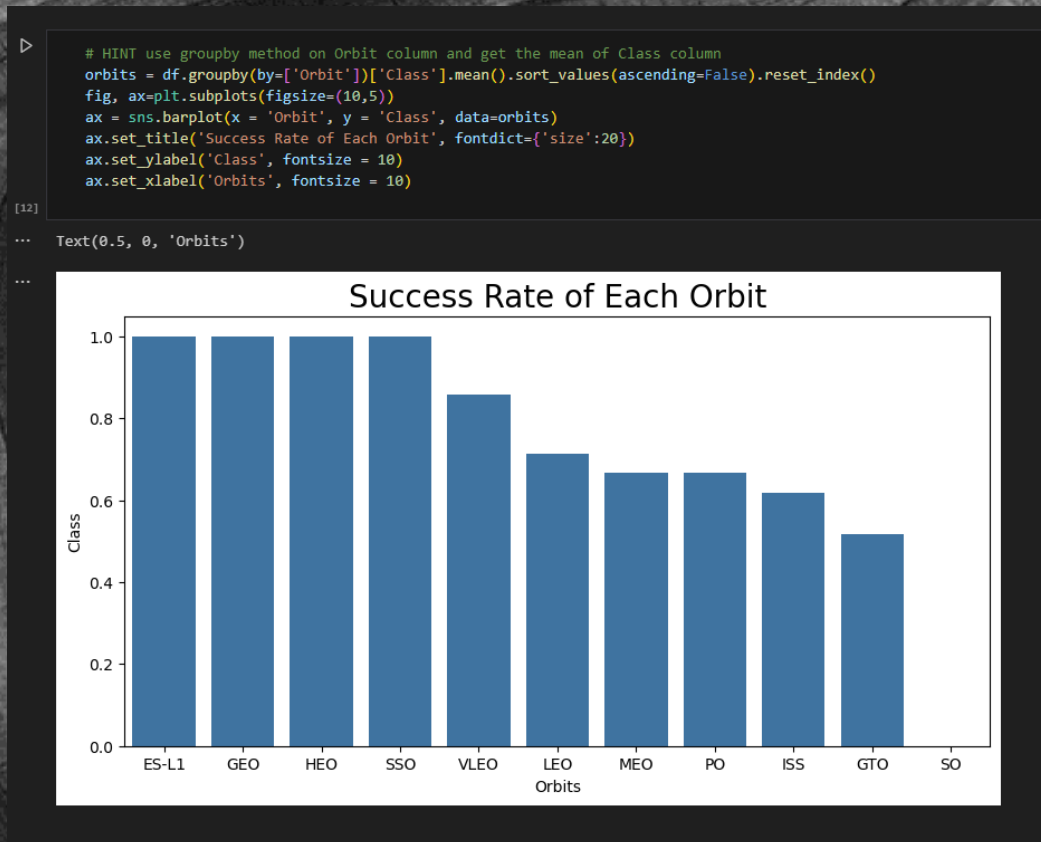


Understand the factors affecting SpaceX launch success.

Improve operational efficiency and reduce costs through data analysis.

Explore historical trends and patterns in launches.

Key questions:



What are the most important factors influencing landing success?

How have success rates evolved over time?

Can the outcome of a launch be accurately predicted?

Data collection:

- SpaceX API for detailed launch information.
- Web scraping of the Falcon9 wiki page to complement the data.
- Data cleaning: Handling of missing values and outliers.
- Standardization of variables such as “outcome” and “landing_outcome”.
- Feature Engineering: Creation of new variables such as: Categories for landing outcome (success, failure, loss).
- Count of launches per site and orbit.



Exploratory Analysis with Visualizations

Key Visualizations:



Bar charts to show the number of launches per site.

Scatter plots to relate payload mass and landing outcome.

Trend curves to observe the evolution of success rates over time.

Relevant results:

Sites such as Cape Canaveral present higher number of successful launches.

GEO and GTO orbits have higher success rates than low orbits.

Exploratory analysis with SQL

1. Obtaining unique launches per site.
2. Count of launches with “CCA” result.
3. Calculation of average payload mass for each orbit.
4. Identification of the number of mission successes and failures.

Insights:

Cape Canaveral site is the most used site for successful launches.

GEO and GTO orbits have higher average payload mass.



Interactive Map with Folium

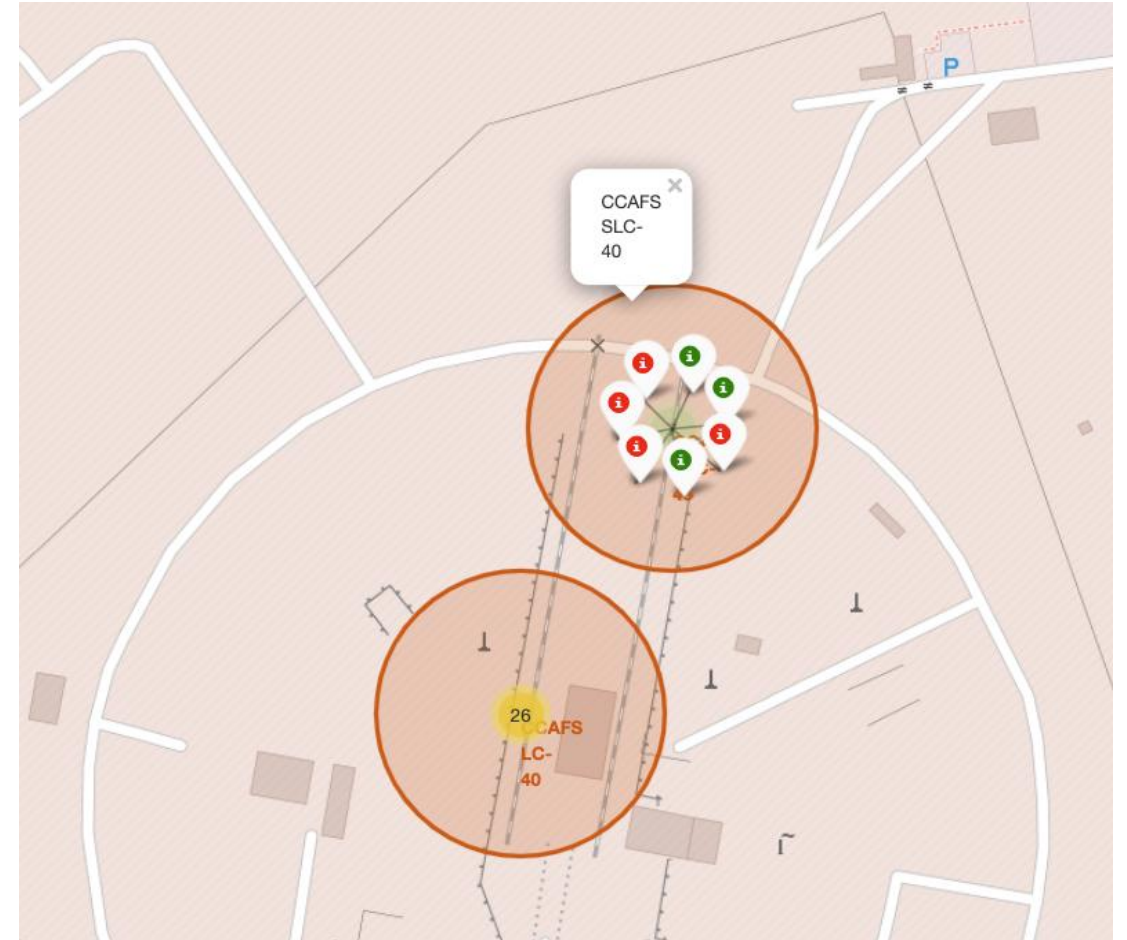
Map features:

Geographic markers to visualize launch sites.

Different colors to distinguish between successes and failures.

Results:

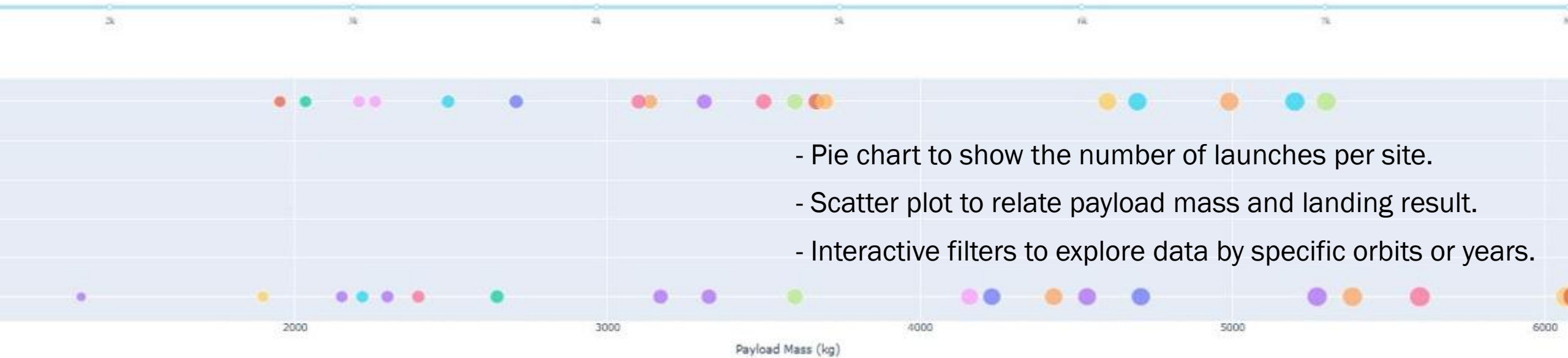
Spatial visualization of the distribution of successful and unsuccessful launches.



Interactive Dashboard with Plotly and Dash



Dashboard components:



- Pie chart to show the number of launches per site.
- Scatter plot to relate payload mass and landing result.
- Interactive filters to explore data by specific orbits or years.

Predictive Analysis

- Model process:
- Division of data into training and testing.
- Use of GridSearchCV to select the best hyperparameters.
- Evaluation using metrics such as accuracy.

```
svm_cv = GridSearchCV(svm, parameters, cv=10, verbose=10)
svm_cv.fit(X_train, Y_train)

[21]
... Fitting 10 folds for each of 75 candidates, totalling 750 fits
[CV 1/10; 1/75] START C=0.001, gamma=0.001, kernel=linear.....
[CV 1/10; 1/75] END C=0.001, gamma=0.001, kernel=linear; score=0.625 total time= 0.8s
[CV 2/10; 1/75] START C=0.001, gamma=0.001, kernel=linear.....
[CV 2/10; 1/75] END C=0.001, gamma=0.001, kernel=linear; score=0.625 total time= 0.5s
[CV 3/10; 1/75] START C=0.001, gamma=0.001, kernel=linear.....
[CV 3/10; 1/75] END C=0.001, gamma=0.001, kernel=linear; score=0.571 total time= 0.2s
[CV 4/10; 1/75] START C=0.001, gamma=0.001, kernel=linear.....
[CV 4/10; 1/75] END C=0.001, gamma=0.001, kernel=linear; score=0.571 total time= 0.5s
[CV 5/10; 1/75] START C=0.001, gamma=0.001, kernel=linear.....
[CV 5/10; 1/75] END C=0.001, gamma=0.001, kernel=linear; score=0.714 total time= 0.0s
[CV 6/10; 1/75] START C=0.001, gamma=0.001, kernel=linear.....
[CV 6/10; 1/75] END C=0.001, gamma=0.001, kernel=linear; score=0.571 total time= 0.0s
[CV 7/10; 1/75] START C=0.001, gamma=0.001, kernel=linear.....
[CV 7/10; 1/75] END C=0.001, gamma=0.001, kernel=linear; score=0.429 total time= 0.0s
[CV 8/10; 1/75] START C=0.001, gamma=0.001, kernel=linear.....
[CV 8/10; 1/75] END C=0.001, gamma=0.001, kernel=linear; score=0.714 total time= 0.0s
[CV 9/10; 1/75] START C=0.001, gamma=0.001, kernel=linear.....
[CV 9/10; 1/75] END C=0.001, gamma=0.001, kernel=linear; score=0.571 total time= 0.0s
[CV 10/10; 1/75] START C=0.001, gamma=0.001, kernel=linear.....
[CV 10/10; 1/75] END C=0.001, gamma=0.001, kernel=linear; score=0.714 total time= 0.0s
[CV 1/10; 2/75] START C=0.001, gamma=0.001, kernel=rbf.....
[CV 1/10; 2/75] END C=0.001, gamma=0.001, kernel=rbf; score=0.625 total time= 0.0s
[CV 2/10; 2/75] START C=0.001, gamma=0.001, kernel=rbf.....
[CV 2/10; 2/75] END C=0.001, gamma=0.001, kernel=rbf; score=0.625 total time= 0.0s
...
```

- Key results:
- The model achieved 90% accuracy in predicting results.
- The most important features are payload mass, orbit and launch site.

Summary of findings:

- Sites such as Cape Canaveral are key to launch success.
- GEO and GTO orbits have higher success rates.
- The predictive model can accurately predict the outcome of launches.
- **Recommendations:**
 - Optimize resources at sites with higher probability of success.
 - Investigate underlying causes of failures in specific orbits.



Thank you

<https://github.com/4sur>
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